

$$R = 795 \, \Omega$$

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C

$$B_S = \mu_0 n I = 4\pi \times 10^{-7} \times 15 \times 0.20$$

$$B_C = \frac{\mu_0 N I}{2r} = \frac{4\pi \times 10^{-7} \times 0.90}{2 \times 1.6}$$

$$B_{\text{resultant}} = B_S - B_C = 3.4 \times 10^{-6}$$

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B

For parallel wires with current flowing in the opposite directions, the wires will repel each